Tanmay Bhuskute

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SUMMARY

Infrastructure and Software Engineer with 2 years operating large-scale cloud distributed systems on AWS at Siemens. Built Kubernetes platforms, CI/CD automation, and Spark data pipelines with Java, Python, Helm, and Terraform; strengths in toil elimination, cost optimization, observability, and applying ML/GenAI to operational efficiency.

EDUCATION

Master of Science in Computer Science Aug 2024 - Dec 2026
University of Cincinnati
Relevant Coursework: Advanced Algorithms, Intelligent Data Analysis, Database Theory.

Bachelor of Engineering in Computer Engineering Aug 2018 - Jul 2022
AISSMS College of Engineering
Relevant Coursework: Data Mining, Data Analytics, Object-Oriented Programming.

TECHNICAL SKILLS

Programming: Java, Python, Go, C/C++, SQL, JavaScript, TypeScript, Bash, functional and object-oriented design.

Data Systems: Apache Spark, PySpark, Delta Lake, medallion pipelines, streaming & batch data processing, ETL, REST APIs.

Infrastructure & Cloud: AWS (EC2, S3, EFS, Lambda, SQS), multi-cloud (Azure, GCP), Kubernetes, ArgoCD, Helm, CRDs, Terraform, Ansible, Vault.

Reliability & Observability: GitLab CI/CD, Jenkins, Docker, Linux, CloudWatch, Datadog, SonarQube, Blue-Green deploys.

ML & GenAI for Ops: Gemini, Vertex AI, LangChain, PyTorch, scikit-learn, pandas, NumPy.

WORK EXPERIENCE

Software Development Engineer | Siemens Industry Software Jul 2022 – Jun 2024

- Operated **40+ large-scale production environments** on AWS, executing **44 upgrades** in 4 months and troubleshooting live distributed-system issues across EC2, Linux, databases, and containerized services.
- Built internal automation frameworks in Python (Boto3) that replaced legacy Ruby scripts, eliminating deployment toil and cutting deployment time by **40%** across AWS environments.
- Migrated **40+ environments** from process-based deployments to **Kubernetes** using ArgoCD, Helm, CRDs, GitLab CI/CD, Amazon EFS, and HashiCorp Vault, driving adoption of scalable deployment patterns.
- Designed **Ansible Blue-Green deployment** framework with automated rollback, reducing production upgrade downtime by **50%** and enabling full recovery within 15 minutes.
- Cut monthly AWS spend **40%** ($15K to $9K) via EC2 rightsizing and automated cleanup of orphaned resources, using CloudWatch and Datadog observability to drive cost and reliability decisions; also raised automated code coverage from **35% to 78%** as SonarQube Specialist.

Graduate Researcher | University of Cincinnati Feb 2026 – Present

- Designed two-stage memory-augmented **LLM pipeline** applying GenAI (Gemini, Vertex AI) to cut operational latency **25.7x** (39.4s vs 1012.1s per run) while preserving task fidelity across 53 runs on 20+ apps.

Software Developer Intern | Siemens Industry Software Mar 2022 – Jun 2022

- Automated AWS provisioning with **Terraform** as Infrastructure as Code (**30%** faster deploys) and built a **Go + Terratest** framework catching **95%** of misconfigurations before production.

Software Developer Intern | Government of Maharashtra, Water Resource Department Sep 2021 – Dec 2021

- Built and deployed a full-stack **React and MySQL** workflow application that digitized departmental record management and improved operational efficiency by **40%**.

PROJECTS AND RESEARCH

NYC Mobility Forecasting [PySpark, Apache Spark, Delta Lake, FastAPI, Redis] Jul 2026

- Built Spark-based **medallion (bronze/silver/gold) pipeline** for large-scale batch data processing of a **2.3M-row** hourly dataset across **263 zones**; served predictions via a FastAPI service with PostgreSQL and Redis caching, containerized with Docker.
- Designed two-stage Poisson hurdle model improving weighted MAPE **18.77% to 16.52%** and cutting RMSE **43%**, MAE **53%** over baseline Random Forest.

Spotify Analytics Platform [Python, FastAPI, PostgreSQL, ETL, Materialized Views] Jul 2026

- Engineered a 4-stage Python **ETL pipeline** for **71K+ streaming events** with rate limiting and backoff retry; designed a PostgreSQL schema with materialized views and 16 SQL RPC functions for low-latency aggregated queries.

EXTRACURRICULAR ACTIVITIES

- **Finalist, RevolutionUC 2025 Hackathon** (300 participants, 31 schools) | **SAFe Practitioner 5.1** (Mar 2023).